22/9/2026

Mid এর আগে
Machine — DC Generator, Motor
electronics — Operational Amplifier

DC Generator

Faraday's laws (২টি)

Slide 2

Slide 4

$e \propto -\frac{d\phi}{dt}$ ← Farady's second law

$e = -N\frac{d\phi}{dt}$

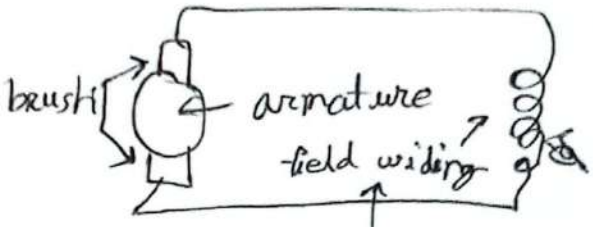
e = induced e.m.f
$\frac{d\phi}{dt}$ = rate of change of flux
N = number of turns of the coil

Lenz law

Slide 6

Generator AC generate করে
কোন generator directly DC generate করতে পারে না।
Split Ring — AC কে DC করতে ← commutator বলা হয়
Primary component
force
magnatic flux
current

ফ্যারাডের Law
লেঞ্জের Law



parallel এ shunt
Series এ series

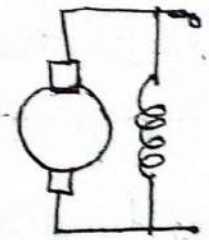
Slide 11

Types of DC Generator

Circuit draw করা Mandatory

Assume 1V voltage drop per brush এবং 2টা brush
∴ Total 2V voltage drop

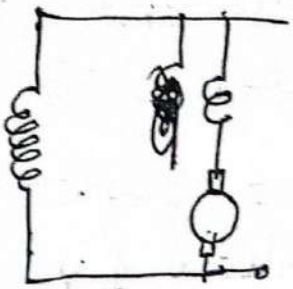
Shunt



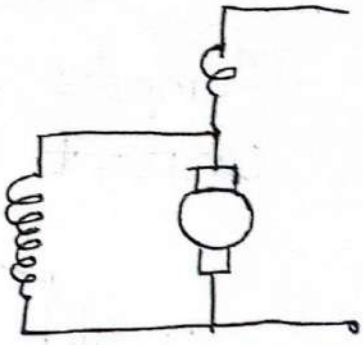
Series



Long shunt compound



Short shunt compound



Lecture 14

Math 1

Example: A shunt generator delevers 450 A at 230 V and the resistance of the shunt field and armature are 50 Ω and 0.03 Ω respectively. Calculate the generated e.m.f.

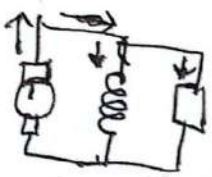
Solution

এটি slide এ আগে আছে

$$I_{sh} = \frac{V}{R} = \frac{230}{50} = 4.6\,A$$

$$I_a = I + I_{sh}$$

$$= 450 + 4.6 = 454.6\,A$$



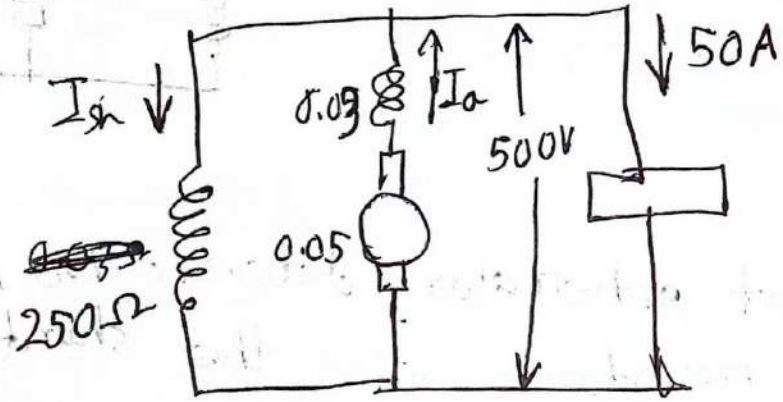
Voltage drop $= I_a R_a = 454.6 \times 0.03 = 13.6\,V$

$$E_g = V_a + I_a R_a = 230 + 13.6 = 243.6\,V$$

28/9/26

Lecture_DC-generator-1.pdf

[1] A long shunt compound generator delivers a load current of 50 A at 500 V and has armature, series field and shunt field resistances are 0.05 Ω, 0.03 Ω and 250 Ω respectively. Compute the generated voltage and armature current. Allow 1 V per brush for contact drop.



$$I_{sh} = \frac{500}{250} = 2A$$

$$I_a = I_{sh} + I = 2 + 50 = 52A$$

Voltage drop $= 0.03 \times 52 + 0.05 \times 2$

$= 6.76 V = 1$

$E_g = 500 + 6.76 = 506.76 V$

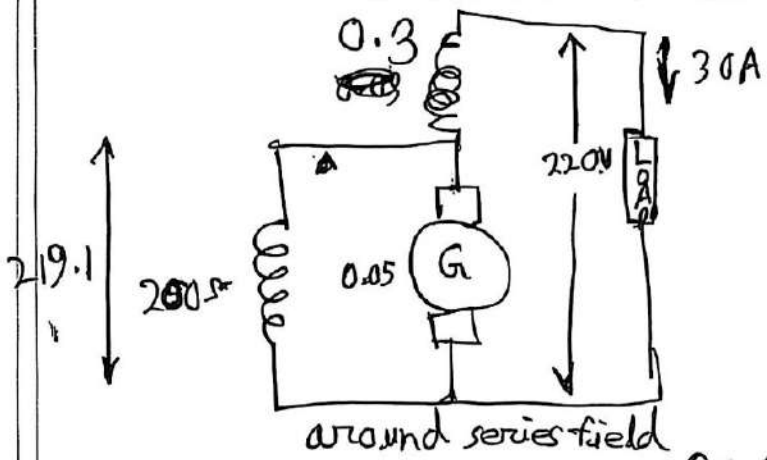
Voltage drop on series $= 52 \times 0.03 = 1.56 V$

armature drop $= 0.05 \times 52 = 2.6$

Drop at brushes $= 2 \times 1 = 2V$

$E_g = 500 + 2.6 + 1.56 + 2 = 506.16 V$

A short shunt compound generator delivers a load current of 30 A at 220 V and has armature, Series field and shunt field resistances are 0.05 Ω, 0.3 Ω and 200 Ω respecectively. Compute the generated voltage armature current. Allow 1 v per brush



Voltage drop = 0.3 × 30 A = 9 V

∴ Voltage around armature = 220 + 9 = 229 V ~~219.1~~

$I_{sh} = \frac{229}{200} = 1.145$ A ~~1.0955~~

$I_a = I_{sh} + 30 = 1.145 + 30 = 31.145$ A ~~31.0955~~

$I_a R_a = 31.145 \times 0.05 = 1.56$ V

Brush drop = 2 × 1 = 2 V

DC Generator generated EMF equation

$$E_g = \frac{\phi Z N}{60} \times \frac{P}{A} \text{ volt}$$

Wave winding = Series
Lap winding = parallal

→ wave winding has only two parallal paths (Low current, High voltage)

Lap winding has multiple parallal paths (High current, low voltage).

ϕ = flux/pole in weber
Z = total no. of armature conductor
P = No of generator poles
A = No of parallal paths in armature
N = armature rotation in revolution per minute
E = EMF induced in any parallal path in armature

for wave winding $A = 2$
for lap winding $A = P$

next week Monday class test

A four pole generator having wave wound armature winding has 51 slots each slot containing 20 conductors. Calculate E_g, when driven at 1500 rpm assuming flux per pole is 7 mWb.

~~170.54~~

$P = 4$

$A = 2$

$N = 1500$

$\phi = 7 \times 10^{-3}$

$Z = 51 \times 20 = 1020$

$$E_g = \frac{7 \times 10^{-3} \times 1020 \times 1500}{60} \times \frac{4}{2}$$

~~17.85~~ $= 357$ V

rps থেকে কিভাবে rpm বের করতে হয়

$rpm = 60 \times rps$

29/9/2026

An 8-Pole dc generator has 800 armature conductors and flux of 0.05 Wb/pole. Compute E_g if it is Lap connected and runs at 1200 rpm. What will be the speed at which it is to be drived to produce the same e.m.f if it is wave wound.

$P = 8$

$A = P = 8$

$N = 1200$

$\phi = 0.05 \times 8 = 0.4$

$Z = 800$ 0.05

$$E_g = \frac{0.4 \times 800 \times 1200}{60} \times \frac{8}{8} = 6400\ V$$

$$= 800\ V$$

if wave wound $A = 2$

$$800 = \frac{0.05 \times 800 \times N}{60} \times \frac{8}{2}$$

$$\Rightarrow N = 300\ rpm$$

1 Wb = 10^{-8} Maxwells (M_x)

Wb = $10^{-8} \times M_x$

An 6-pole dc generator has 800 armature conductors and flux of ~~0.05 Wb/pole~~ ~~Compute Eg = 800 V if~~ 27000 ~~Maxwell~~ M_x/Pole

Compute Eg if it is lap connected and runs at 70 r.p.s. What will be the speed at which it is to be drived to produce the same e.m.f if it is wave wound.

$P = 6$

$Z = 800$

$\phi = 27000 \times 10^{-8} = 2.7 \times 10^{-4}$ Wb

$A = P = 6$

$N = 70 \times 60 = 4200$ rpm

$$E_g = \frac{\phi Z N}{60} \times P = \frac{2.7 \times 10^{-4}}{60} \times 800 \times 4200 \times \frac{6}{6}$$

$$= 15.12 \text{ V}$$

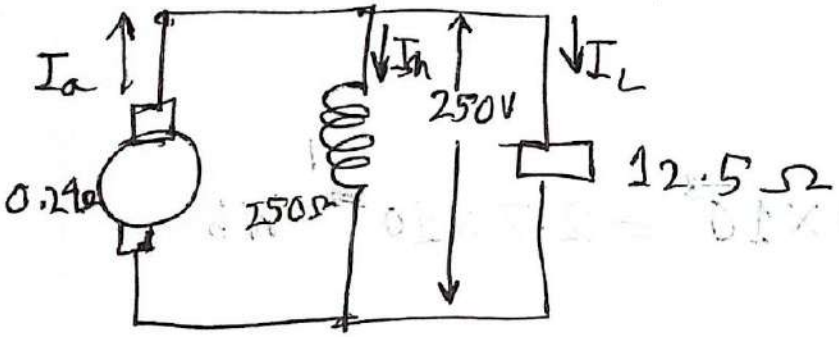
wave wond $A = 2$

$$15.12 = \frac{2.7 \times 10^{-4} \times 800 \times N}{60} \times \frac{6}{2}$$

$\Rightarrow N = 1400$ rpm

An 8 pole, dc Shunt gen with 778 wave Connected armature conductors and running at 500 r.p.m. Supplies a load of 12.5Ω resistance at terminal voltage of 250 v. The armature resistance is 0.24Ω and field resistance is 250Ω. Find armature current, induced e.m.f and flux per pole

↑ 9.83 mwb



$$I_{sh} = \frac{V}{R} = \frac{250}{250} = 1\ A$$

$$I_L = \frac{V}{R} = \frac{250}{12.5} = 20\ A$$

armature current $I_a = I_{sh} + I_L = 1 + 20 = 21\ A$

Voltage drop $= I_a R_a = 21 \times 0.24 = 5.04\ V$

$\therefore$ induced e.m.f $= 250 + 5.04$

$= 255.04\ V$

$E_g = 255.04$

$\phi = ?$

$P = 8$

$Z = 778$

$A = 2$

$N = 500$

$$255.04 = \frac{\phi' \times 778 \times 500}{60} \times \frac{8}{2}$$

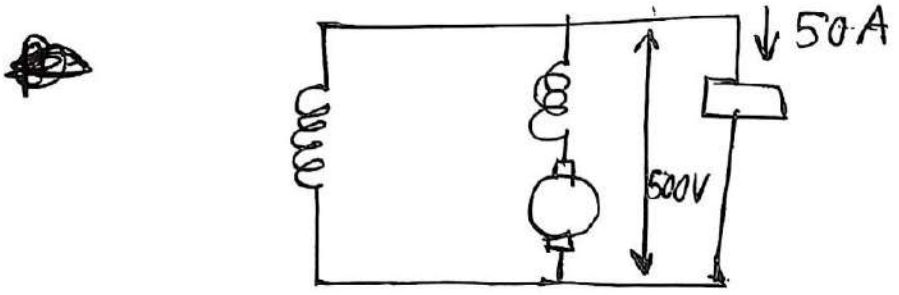
$$\Rightarrow \phi = 9.83 \times 10^{-3} \text{ Wb}$$

A Textbook of electrical technology } Machine related Topic
BL Theraja (Volume-II)

A 4 pole, long shunt, lap wound gen delivers load current, 50 A and voltage 500 V

The armature, series and shunt field resistances are 0.03 Ω, 0.04 Ω, 200 Ω. The brush drop is 1 V/brush. Determine induced voltage.

Calculate the no of conductors if the speed is 1200 rpm and flux per pole is 0.02 wb.



6/10/2026

DC Motor

Lecture 9_ DC Motor

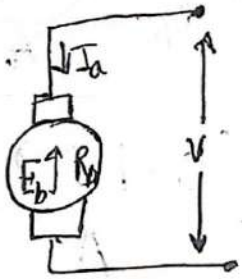
A DC motor is an electrical machine that converts electrical energy into mechanical energy

input – electrical
output - mechanical

Torque:

Torque is the measure of the force that can cause an object to ro_____

Flemings left hand rule (DC generator ✓ Right hand rule)



$V = E_b + I_a R_a$

E_b = back EMF

The direction of the induced emf is opposite to the applied voltage

$$E_b = \frac{NP\phi Z}{60A}$$

E_b = the induced emf of motor
A = the number of parallel paths through the armature
P = The number of poles
N = The speed
Z = The total no of conductor
ϕ = flux per pole

The back emf makes the DC motor, self regulating machine.

Torque and Speed Equation

Torque equation,

We know $V = E_b + I_a R_a$

$$V I_a = E_b I_a + I^2 R_a$$

Mechanical Power required to rotate the shaft $I\omega$; $\omega = \frac{2\pi N}{60}$ rps

So, $T\omega = E_b I_a$

Put the value of $E_b = \frac{N \phi Z P}{A 60}$

$$T = \left(\frac{0.159 P Z}{A}\right) \phi I_a$$

$\frac{1}{2\pi} = 0.159$

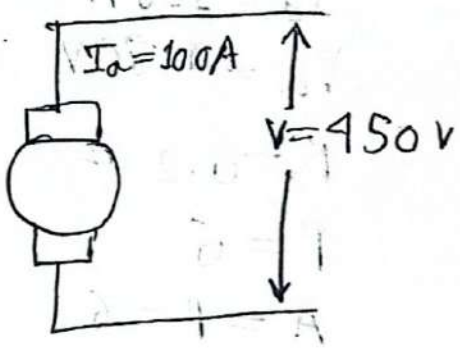
T = Torque (τ)

Slide 15/23

1. $I_a = 100A$

$V_a = 450V$

$R_a = 2.5\Omega$



$P = 6$

$A = 2$

$Z = 850$

$\phi = 0.05Wb$

$V = E_b + I_a R_a$

$450 = E_b + 100 \times 2.5$

$\Rightarrow E_b = 450 - 100 \times 2.5$

$= 200\ V$

$$E_b = \frac{N P \phi Z}{60 A}$$

$$200 = \frac{N \times 6 \times 0.05 \times 850}{60 \times 2}$$

$$\Rightarrow N = \frac{200 \times 60 \times 2}{6 \times 0.05 \times 850} = 94.12$$

$$T = \left(\frac{0.159 \times 6 \times 850}{2}\right) \times 0.05 \times 100$$

$$= 2027.25\ Nm$$

2. $I_a = 110\ A$

$V_a = 480\ V$

$R_a = 0.2\ \Omega$

$P = 6$

$A = P = 6$

$Z = 864$

$\phi = 0.05\ Wb$

$$V = E_b + I_a R_a$$

$$\Rightarrow 480 = E_b + 110 \times 0.2$$

$$\Rightarrow E_b = 480 - 110 \times 0.2 = 458\ V$$

$$E_b = \frac{NP\phi Z}{60A}$$

$$\Rightarrow 458 = \frac{N \times 6 \times 0.05 \times 864}{60 \times 6}$$

$$\Rightarrow N = \frac{458 \times 60 \times 6}{6 \times 0.05 \times 864} = 636.11$$

$$T = \left(\frac{0.159 PZ}{A}\right) \phi I_a$$

$$= \frac{0.159 \times}{}$$

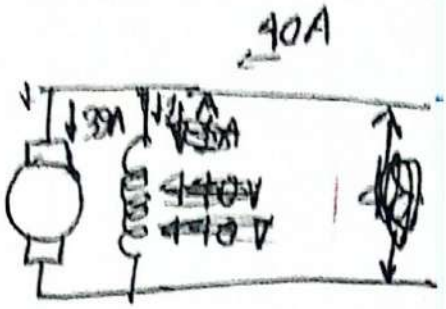
$P = 6$

$Z = 540$

$A = P = 6$

$\phi = 30\ mWb = 0.03\ Wb$

$R_a = 0.05\Omega$



$P_a = 8.595kW = 8595\ W$

$P_a = \frac{V_a^2}{R_a}$

$\Rightarrow V_a = \sqrt{P_a R_a} = \sqrt{8595 \times 0.05} = 20.73V$

~~$V_a = 20.73$~~
~~$R_a = 0.05$~~

$I_a = 39\ A$

$V = E_b + I_a R_a$

$20.73 = E_b + 39 \times 0.05$

$\Rightarrow E_b = 0.02667$

$E_b = \frac{NP\phi Z}{60A}$

$\Rightarrow 0.02667 = \frac{N \times 6 \times 0.03 \times 540}{60 \times 6}$

$\Rightarrow N = \frac{0.02667 \times 60 \times 6}{6 \times 0.03 \times 540} = 0.988$

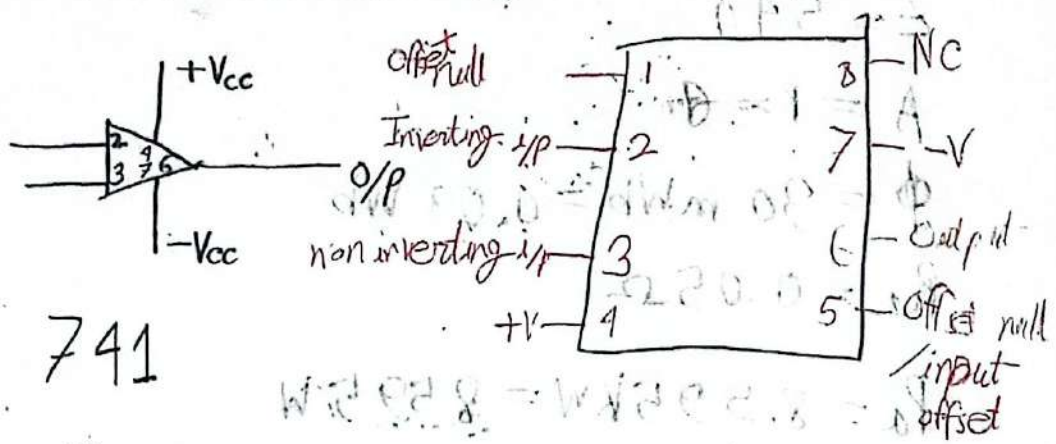
~~$\frac{NP\phi Z}{60A}$~~ $T = \left(\frac{0.159PZ}{A}\right)\phi I_a$

$= \frac{0.159 \times 6 \times 540}{6} \times 0.03 \times 39 = 37.276$

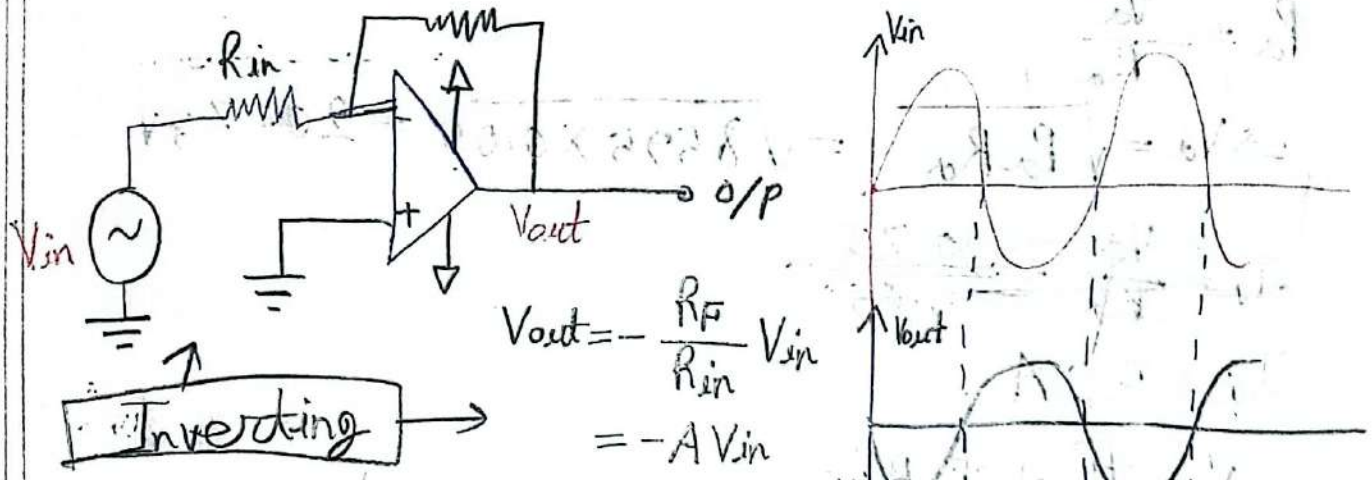
2/10/26

Mice sound ক Amplify করতে

Operational Amplifier



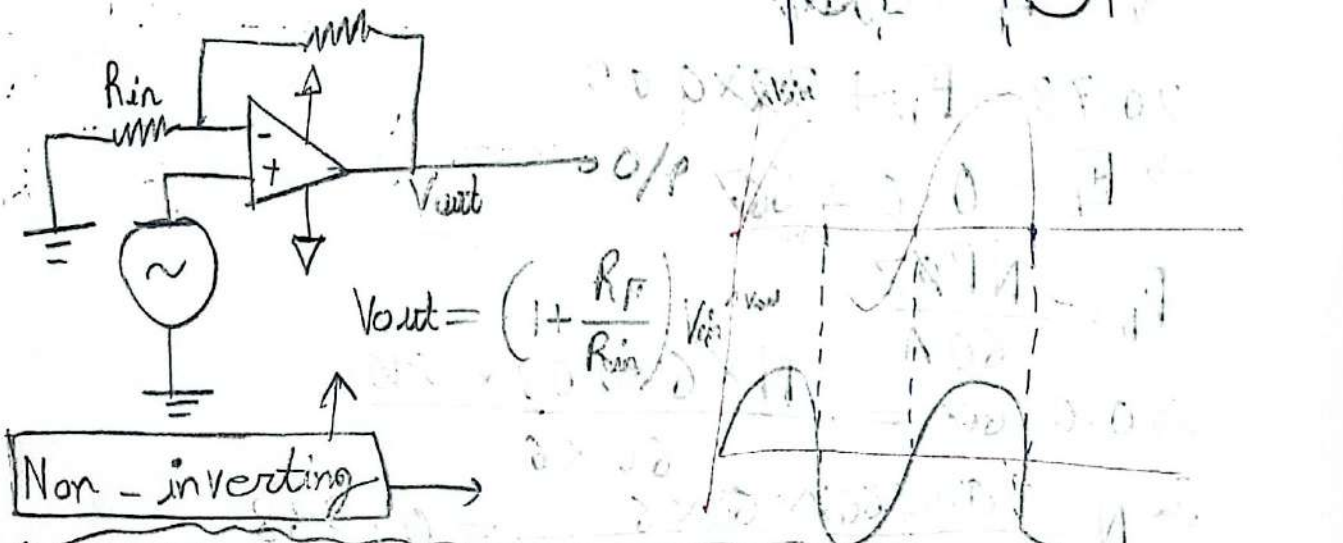
741



$$V_{out} = -\frac{R_F}{R_{in}} V_{in}$$

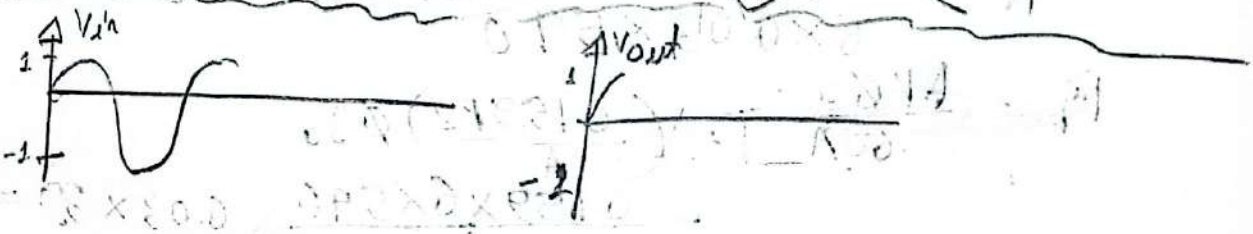
$$= -A V_{in}$$

Inverting →

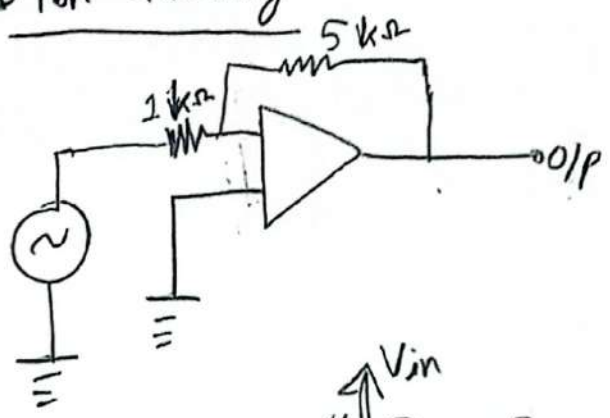


$$V_{out} = \left(1 + \frac{R_F}{R_{in}}\right) V_{in}$$

Non - inverting →



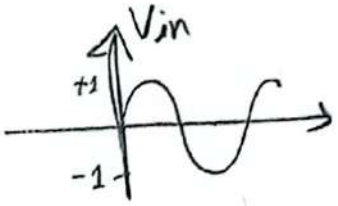
Non-inverting

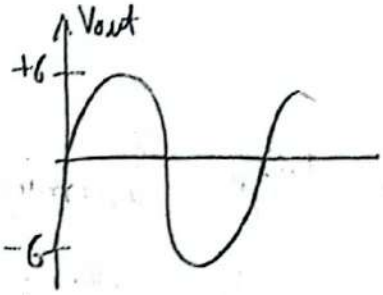


$$V_{out} = \left(1 + \frac{R_F}{R_{in}}\right) V_{in}$$

$$= \left(1 + \frac{5}{1}\right) \times 1$$

$$= 6\ V$$



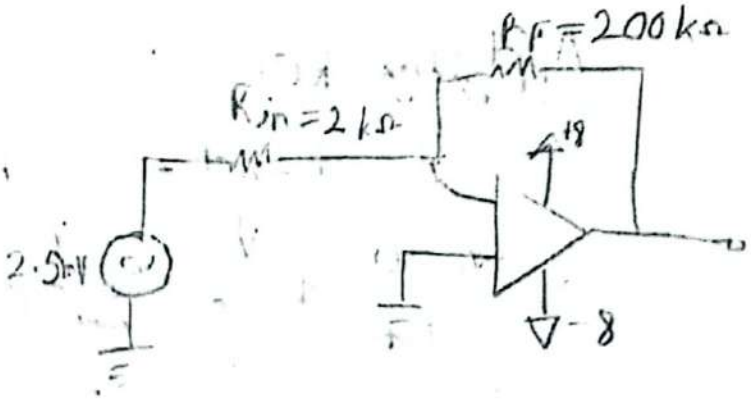


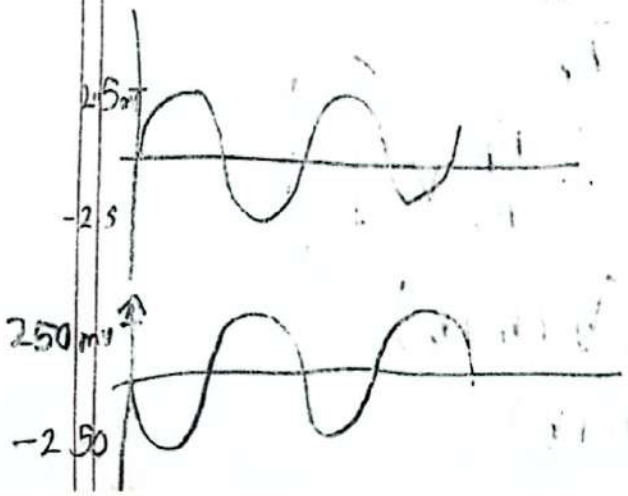
Inverting

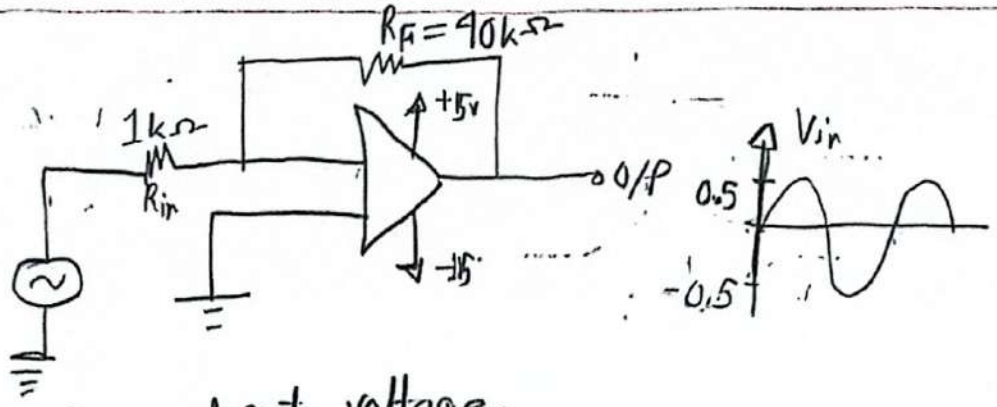
$$V_{out} = -\frac{R_F}{R_{in}} \times V_{in}$$

$$= -\frac{200}{2} \times 2.5$$

$$= -250\ mV$$



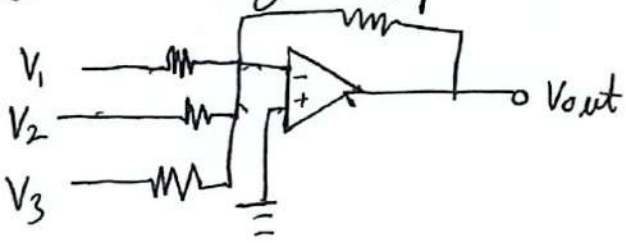




Determin output voltage.

$$V_{out} = -\frac{R_F}{R_{in}} \times V_{in} = -\frac{40}{1} \times 0.5 = -20\,V$$

Inverting Summing amplifier



Applying KCL,

$$I_1 + I_2 + I_3 = I_F$$

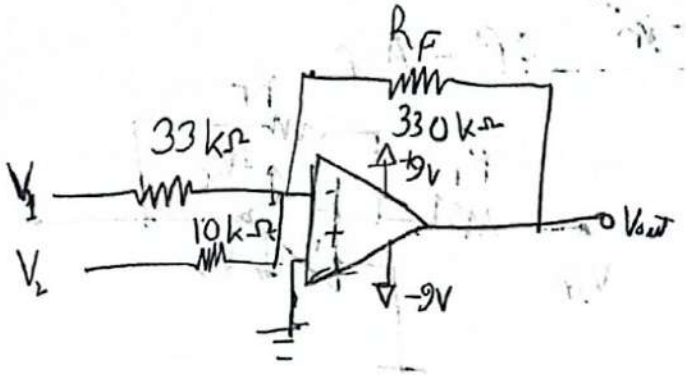
$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_F}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_F}$$

$$\Rightarrow V_{out} = -\left[\frac{R_F}{R_1} V_1 + \frac{R_F}{R_2} V_2 + \frac{R_F}{R_3} V_3\right]$$

If, $R_1 = R_2 = R_3$, $V_{out} = -\frac{R_F}{R}(V_1 + V_2 + V_3)$

If $R_F = R$, $V_{out} = -(V_1 + V_2 + V_3)$



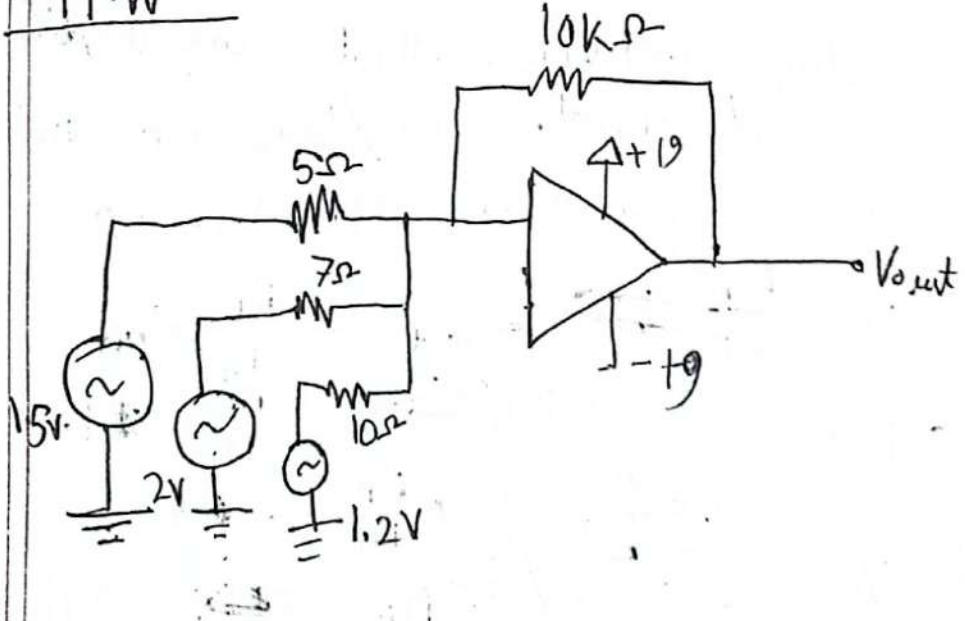
$v_1 = 50\sin(100t)$ mV

$v_2 = 10\sin(300t)$ mV

$$V_{out} = -\left[\frac{330}{33} \times 50\sin(100t) + \frac{330}{10} \times 10\sin(300t)\right]$$

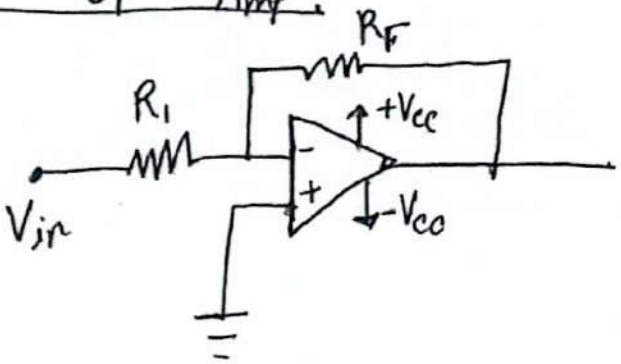
$$= -1500\sin(100t) - 330\sin(300t) \text{ mV}$$

H.W



Virtual ground in op-Amp.

$$A_V = \frac{V_{out}}{V_{in}}$$



$$V_{out} = A_V \times V_{in}$$

$$= A_V \times V_d$$

$$5 = 10^5 \times V_d$$

$$V_d = 0.00005$$

$$V_d \approx 0$$

$$V^+ - V^- = 0; \quad V^+ = V^-$$

i/p = input
o/p = output

Develop an inverting amplifier that can amplify 5 times of input signal. Show/draw i/p and o/p waveforms. Consider a 3V(P-P) as i/p.

Ans:

$$V_{out} = -\frac{R_F}{R_1} V_{in}$$

$$= -\frac{10k\Omega}{2k\Omega} \times 1.5$$

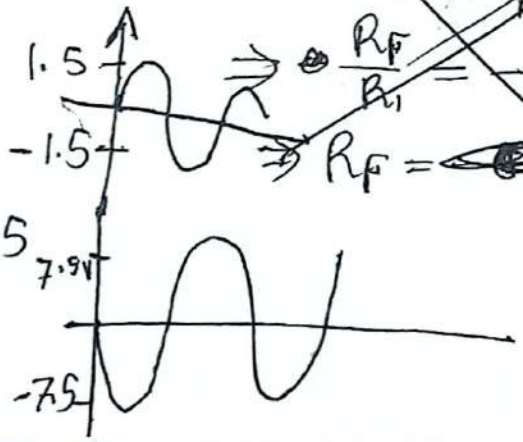
$$= -7.5V$$

$$V_{out} = -\frac{R_F}{R_1} \times V_{in}$$

$$1.5 \times 5 = -\frac{R_F}{R_1} \times 1.5$$

$$\frac{R_F}{R_1} = -5$$

$$R_F = -5R_1$$

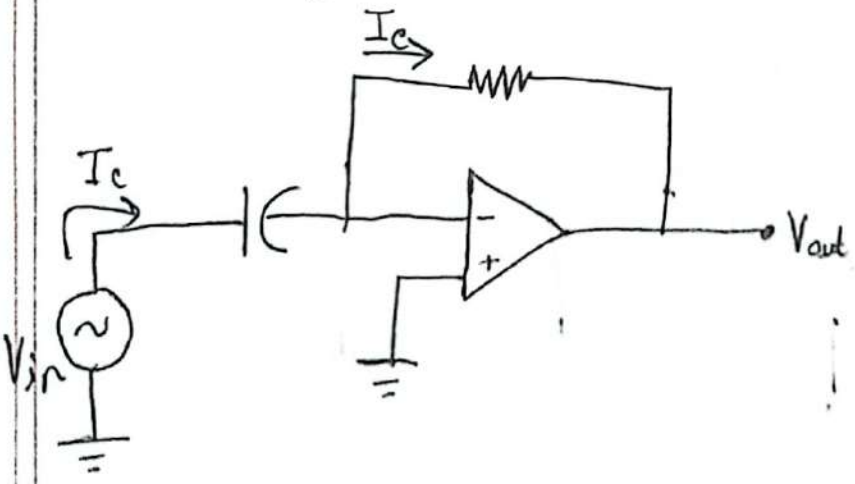


Slew rate

Slew rate is defined as the maximum rate of change of o/p voltage per unit time (V/s). How fast the op-amp is able to reach its final value / How fast the o/p voltage can change

OP-amp differentiator

$V_{in}(t) \rightarrow$ [Differentiator] $\rightarrow V_{out} = \frac{dV_{in}}{dt}$

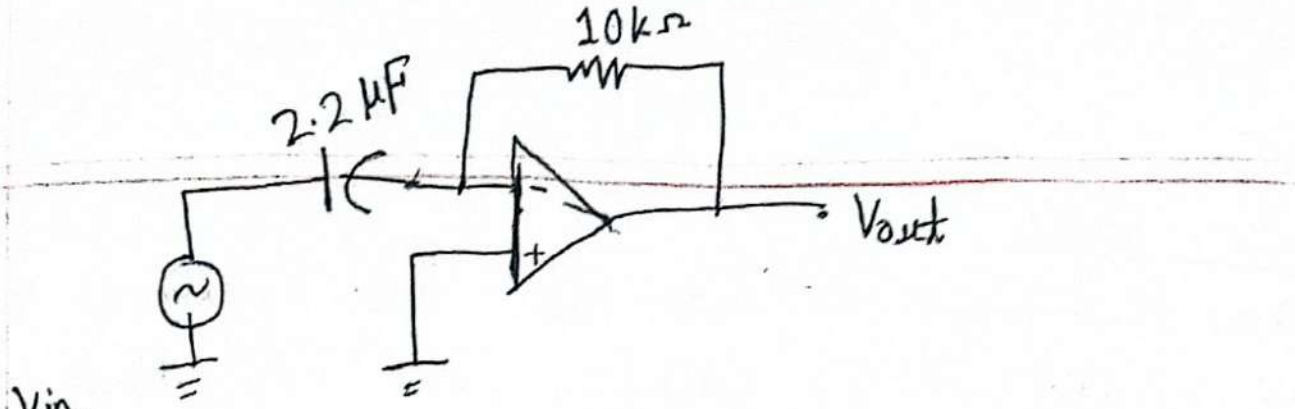


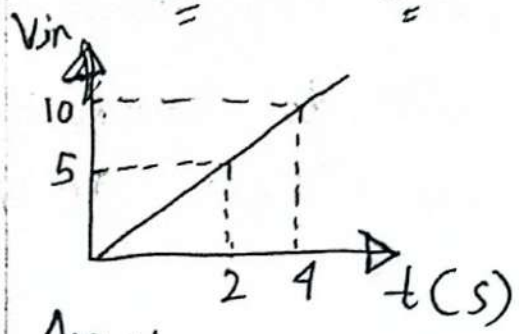
$I_c = I_f$

$\Rightarrow C\frac{dV_c}{dt} = \frac{0 - V_{out}}{R_F}$

$\Rightarrow C\frac{dV_{in}}{dt} = -\frac{V_{out}}{R_F}$

$\Rightarrow V_{out} = -R_F C\frac{dV_{in}}{dt}$





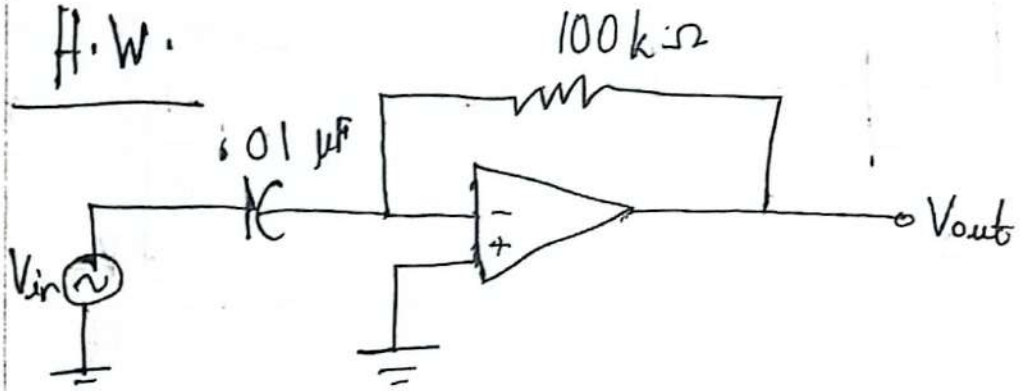
Ans:

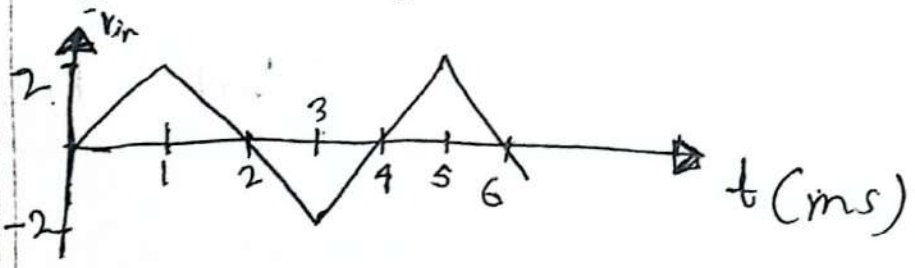
$$V_{out} = -R_F C \frac{dV_{in}}{dt}$$

$$\Rightarrow V_{out} = -10000 \times 2.2 \times 10 \times 10^{-6} \times \frac{5-0}{2-0}$$

$$= -0.55\ V$$

H.W.





Plot the o/p waveform of the op-amp differentiator given ~~below~~ up.